Scalability & Future Plan

Date	26 October 2026
Team ID	EN25CS3011092
Project Name	Global Olympic Sports Performance Analysis Dashboard (Tableau)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static CSV dataset with no live data refresh.	New Games (2028 onwards) and updated results are not reflected, so insights become outdated.	High
2	The athletes dataset has no year column.	The Year filter cannot change athlete-based charts, which limits time analysis of athletes.	Medium
3	60 countries and 21 sport categories make some charts and legends crowded.	Harder to read quickly on small screens.	Medium
4	No forecasting or predictive analysis.	Cannot project future medal counts or efficiency trends.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Workbook published on Tableau Public and a local Flask web app.	Host the Flask app on a cloud service and use Tableau Cloud / Server for many concurrent viewers with role-based access.
2	Data Storage	Static CSV files packaged in the workbook.	Move to a cloud database (SQL / data warehouse) with scheduled extract refresh.
3	Performance	Smooth on about 9,900 rows across the four files.	Use extracts, aggregated data and context filters to handle millions of rows.
4	Security	Public data, no access control.	Add user authentication to the web app, row-level security and permission groups on Tableau Server.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Add new Games (2028 onwards) and more years of data; connect a live data source with automatic refresh.	1-2 months	Always up-to-date insights for decision makers.
Phase 3	Add medal forecasting models and what-if parameters for team size and sports investment.	3-4 months	Supports future planning and funding decisions.
Phase 4	Publish web / mobile dashboards with alerts; add athlete-level yearly data, population and GDP data.	6 months	Wider access for committees, researchers and journalists.

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